Oceanus Folk Explorer

Visual Analysis of Musical Influence (VAST 2025, Mini-Challenge 1)

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Abstract

Oceanus Folk Explorer is a web tool for exploring a large music knowledge graph. The graph describes artists, songs, albums, labels, and the ways works influence one another. The tool helps an analyst answer three questions: who shaped the career of the star Sailor Shift, how the Oceanus Folk genre spread over time, and which artists are likely to rise next. It uses three linked views built with D3.js: an influence network placed on a time axis, a stacked-area chart of genre spread, and a trajectory chart with a ranked list of rising stars. Selections and a time filter are shared across the views.

1 Introduction

The data comes from a fictional music scene on Oceanus Island. The central figure is Sailor Shift, a singer who made the "Oceanus Folk" genre popular. The data is stored as a knowledge graph: nodes are people, groups, songs, albums, and record labels, and edges describe roles (who performed or wrote a work) and influence (which work drew from which).

An analyst working with this data has three goals:

1. Profile Sailor Shift: who influenced her over time, who she influenced, and her impact on the Oceanus Folk community.
2. Understand the genre: how Oceanus Folk spread to other genres, whether the rise was steady or came in bursts, and which genres it drew from.
3. Find rising stars: define what makes an artist promising, compare careers, and point to the next likely stars.

The graph is large, so drawing all of it at once produces an unreadable tangle of lines, often called a "hairball". The tool avoids this in two ways. It focuses on the part of the graph around Sailor Shift, and it uses time as a strong layout rule so the picture stays ordered. The interface is built with Vue 3, the charts with D3.js (version 7), and the linking between views with crossfilter. Heavy computation is done once in Python and exported as small JSON files that the browser loads quickly.

2 The Data

The graph is a directed multigraph with **17,412 nodes** and **37,857 edges**, split into 18 connected components. There are five node types and twelve edge types. Table 1 lists the node types and Table 2 the edges.

Table 1: Node types and their main properties.

Type	Main properties	Count
Person	name, stage name	11,361
Song	name, genre, release year, notable, notoriety year	3,615
RecordLabel	name	1,217
Album	name, genre, release year, notable, notoriety year	996
MusicalGroup	name	223

Table 2: Edge types. Role edges connect people and works; influence edges connect a later work to the earlier work it drew from.

Group	Edge type	Meaning (source → target)
Role	PerformerOf	person/group performed a work
	ComposerOf	person composed a work
	ProducerOf	person/label produced a work or act
	LyricistOf	person wrote lyrics for a work
	MemberOf	person is a member of a group
	RecordedBy	work was recorded by a label
	DistributedBy	work was distributed by a label
Influence	InStyleOf	work performed in the style of the target
	InterpolatesFrom	work reuses a melody from the target
	CoverOf	work is a cover of the target
	LyricalReferenceTo	work refers to the target in its lyrics
	DirectlySamples	work samples the target's audio

Reading direction of influence. For the five influence edges, the *target* is the earlier work that inspired, and the *source* is the later work that drew from it. In other words, the target is the influencer. This single rule lets the tool place influence on a time axis, with older works on the left and newer works on the right.

Time and popularity. Dates are stored as years. A work has a release year, and notable works also have a "notoriety year", which is the year the work first reached a top chart. The notable flag and the notoriety year are the clearest signals of popularity in the data.

Derived data. Influence is recorded between works, not directly between people. To study people, the tool builds person-to-person influence by following the creators of each work: if a work by artist A draws from a work by artist B, then B influenced A. Three small files are produced in Python:

- `sailor_ego.json` – Sailor Shift's influence neighborhood (138 artists and 541 influence links at one hop), with a year and simple metrics for each artist.
- `genre_spread.json` – yearly counts of works influenced by Oceanus Folk, grouped by genre, plus the genres it drew from and the artists it affected.
- `rising_stars.json` – a Rising Star Score and a career trajectory for each Oceanus Folk artist.

The data has a few rough edges that the preprocessing handles. Some influence edges point at a person or a group instead of a work, so both ends of every influence edge are resolved to people before counting. A small number of works have no year and are left out of time-based charts.

3 Related Work

Two classic ways to show a graph are the node-link diagram and the adjacency matrix. Node-link diagrams are easy to read when small but become tangled when large. Matrices stay readable when dense but hide paths. Hybrid designs such as NodeTrix [1] mix both. To keep the focus clear, many tools show one node and its neighborhood, which is the approach used here.

Graphs that change over time need an extra cue for time. A survey by Beck et al. [2] groups these methods into animation and timeline-based layouts. Placing time on an axis, as this tool does for influence, is a timeline-based choice that keeps the order of events visible without animation.

To show how parts of a whole change over the years, the stacked graph, or streamgraph [3], is a common choice and has been used for music listening histories. The genre-spread view uses a stacked area of the same family. For working across several views at once, coordinated multiple views with linked selection and brushing [4] are a standard pattern, and the tool follows it. All charts are drawn with D3 [5].

4 Design Choices

Scope. The tool starts from Sailor Shift and shows her neighborhood, not the whole graph. This keeps the first screen readable and matches the main analytical question.

Layout. In the influence network, the horizontal position of an artist is the year they became active. A force simulation only spreads nodes vertically and keeps them from overlapping, while a strong horizontal force pins each node to its year. The result reads from left to right as a flow of influence through time, instead of a shapeless cloud.

Color. Genre is the main color channel. Oceanus Folk uses a clear cyan so it stands out, and other genres use a categorical palette. Influence edges are colored by their type. In the rising stars view, the five parts of the score each have a fixed color so the same color always means the same thing.

Size and shape. Node size in the network grows with the number of other artists an artist influenced, so important artists are larger. Bars and areas use length and height, which are easy to compare.

Transformations. Career trajectories are cumulative sums, so a steeper line means faster growth. Influence reach is passed through a logarithm so that one very prolific artist does not flatten the scale for everyone else. Score parts are normalized to the range zero to one before they are combined.

Readability. Only the focus artist is labeled in the network. Full details appear in a side panel when a node is selected, which keeps the chart clean while still giving access to the numbers.

5 The Tool

The tool has three views, chosen with a tab at the top. The views share the current selection and the current year filter.

5.1 Artist View

This view answers the first question. The center is the influence network described above. Older

Figure 1: The artist view: influence network on a time axis, a year timeline, a ranking of top influencers, and a detail panel.

influencers sit to the left of Sailor Shift and the artists she influenced sit to the right. A timeline below the network shows how many artists were active each year and can be brushed to filter the other views by year. A ranking lists the artists who influenced the most others. A detail panel shows the selected artist's genre, active years, number of works, and the lists of who influenced them and who they influenced. Clicking a node, a ranking row, or a name in the panel selects that artist and updates every part of the view.

5.2 Genre Spread View

This view answers the second question. The main chart is a stacked area. For each year it shows how many works that were influenced by Oceanus Folk appeared, split by the genre of those works. A dashed line shows Oceanus Folk's own releases for comparison. The shape of the area answers whether the rise was steady or came in bursts: the data shows clear peaks around 2017, 2020, and a large one in 2023, so the rise was intermittent. Side panels rank the genres Oceanus Folk drew from (Indie Folk is by far the largest, followed by Synthwave and Dream Pop), the genres it influenced (Desert Rock, Indie Folk, Dream Pop, and Space Rock lead), and the artists most affected by it.

5.3 Rising Stars View

This view answers the third question. The left chart compares career trajectories. Each line is an artist's cumulative number of chart hits over time, so a steep recent slope marks a fast riser. Sailor

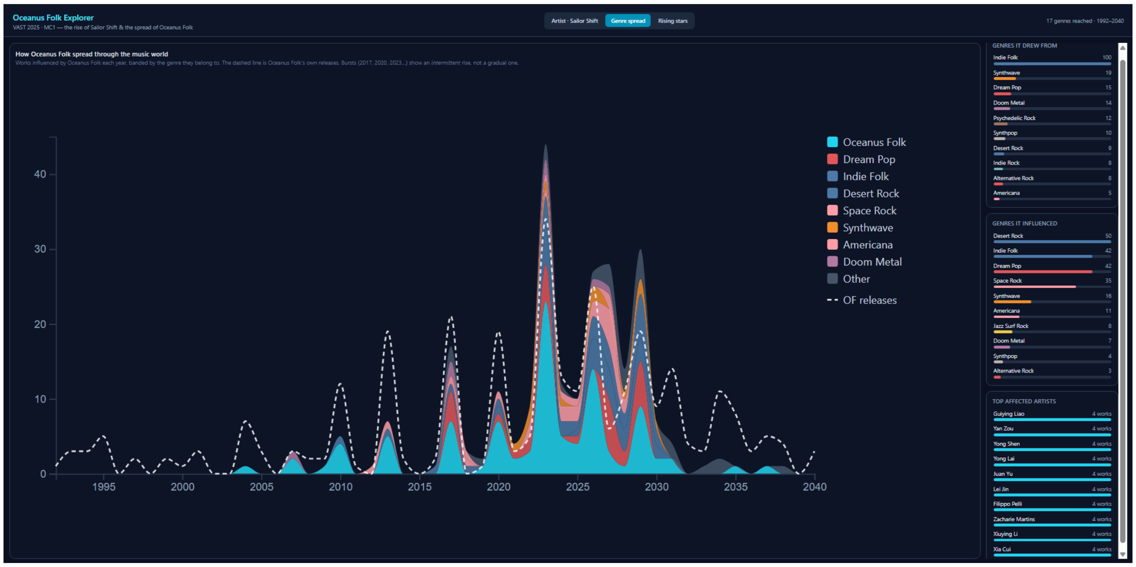

Figure 2: The genre spread view: a stacked area of works influenced by Oceanus Folk over time, with rankings of genres and affected artists.

Shift is drawn as a dashed benchmark line. The right side is a leaderboard of candidates, ranked by a Rising Star Score. The score is the weighted sum of five parts: recent chart hits, momentum (recent hits minus earlier hits), influence reach, share of Oceanus Folk works, and career youth. Each row's bar is split into these five parts in their fixed colors, so the analyst can see why an artist ranks where it does. Clicking a candidate adds or removes its line from the trajectory chart.

5.4 Linking

The views are connected in two ways. Selecting an artist in the artist view updates the network highlight, the ranking, and the detail panel together. Brushing a year range in the timeline filters the network and the ranking through crossfilter. The tab bar switches the analytical task while keeping the same underlying data in memory.

6 Use Case

Suppose an analyst wants to trace Sailor Shift's career and then look ahead. They open the artist view. Sailor Shift sits near 2024. To her left are the artists who influenced her, such as Coralia Bellweather and Barnaby Quill, connected by "in the style of" and "interpolates from" links. To her right are the artists she influenced, such as Bea Azure and Zara Quinn. Selecting Coralia Bellweather shows that she is both an early influence on Sailor and a strong influencer in her own right.

The analyst then switches to the genre spread view. The stacked area shows that Oceanus Folk did not grow smoothly; it grew in bursts, with a large jump in 2023. The side panels show that the genre drew most heavily from Indie Folk and in turn spread into Desert Rock, Dream Pop, and Space Rock. This explains how the genre evolved by borrowing from and feeding back into nearby styles.

Finally, the analyst opens the rising stars view. Sailor Shift's trajectory is the highest, which fits her role as the established star. The leaderboard ranks other Oceanus Folk artists by the Rising

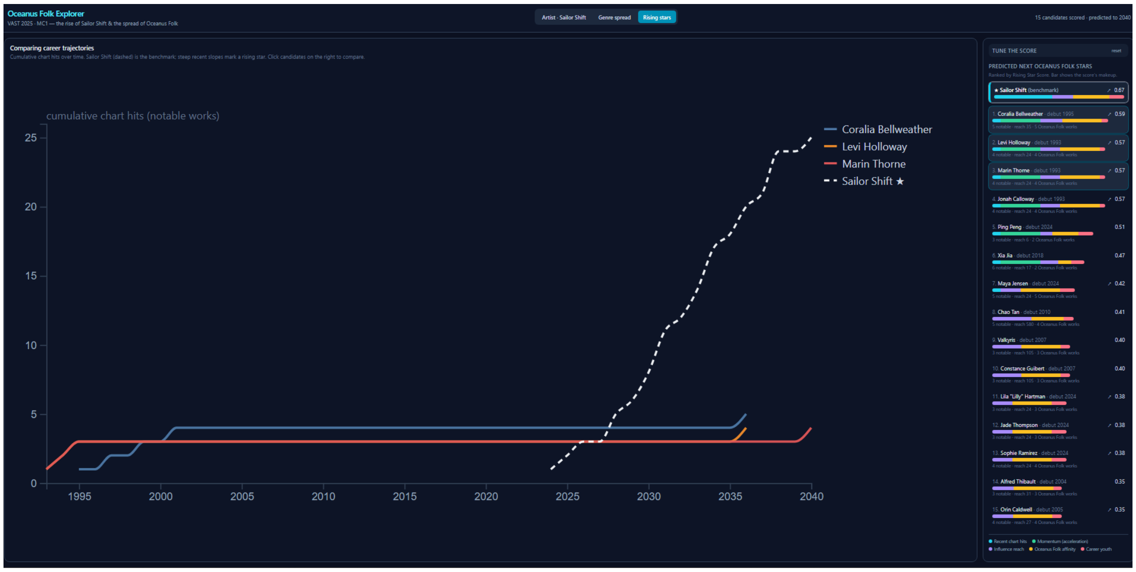

Figure 3: The rising stars view: career trajectories on the left and a ranked leaderboard with a score breakdown on the right.

Star Score, and the breakdown bars show that some are ranked for their recent chart hits while newer artists such as Arlo Sterling rank for youth and momentum. Adding a few of these artists to the trajectory chart lets the analyst compare their growth directly and pick the most promising ones.

7 Conclusion

Oceanus Folk Explorer turns a large and tangled knowledge graph into three readable views that answer concrete questions about influence, genre spread, and rising stars. The main design idea is to use time as the backbone of the layout and to focus on the area around Sailor Shift, which keeps the graph clear.

The tool has limits. The influence edges are sparse and were collected by hand, so some relationships are missing. Works with many creators can inflate influence counts, which the score reduces with a logarithm but does not fully remove. The data also includes future years, so predictions should be read as a guide rather than a fact. Useful next steps are a view for collaborations, controls to change the score weights, and links that carry a selected artist from one view to another.

References

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